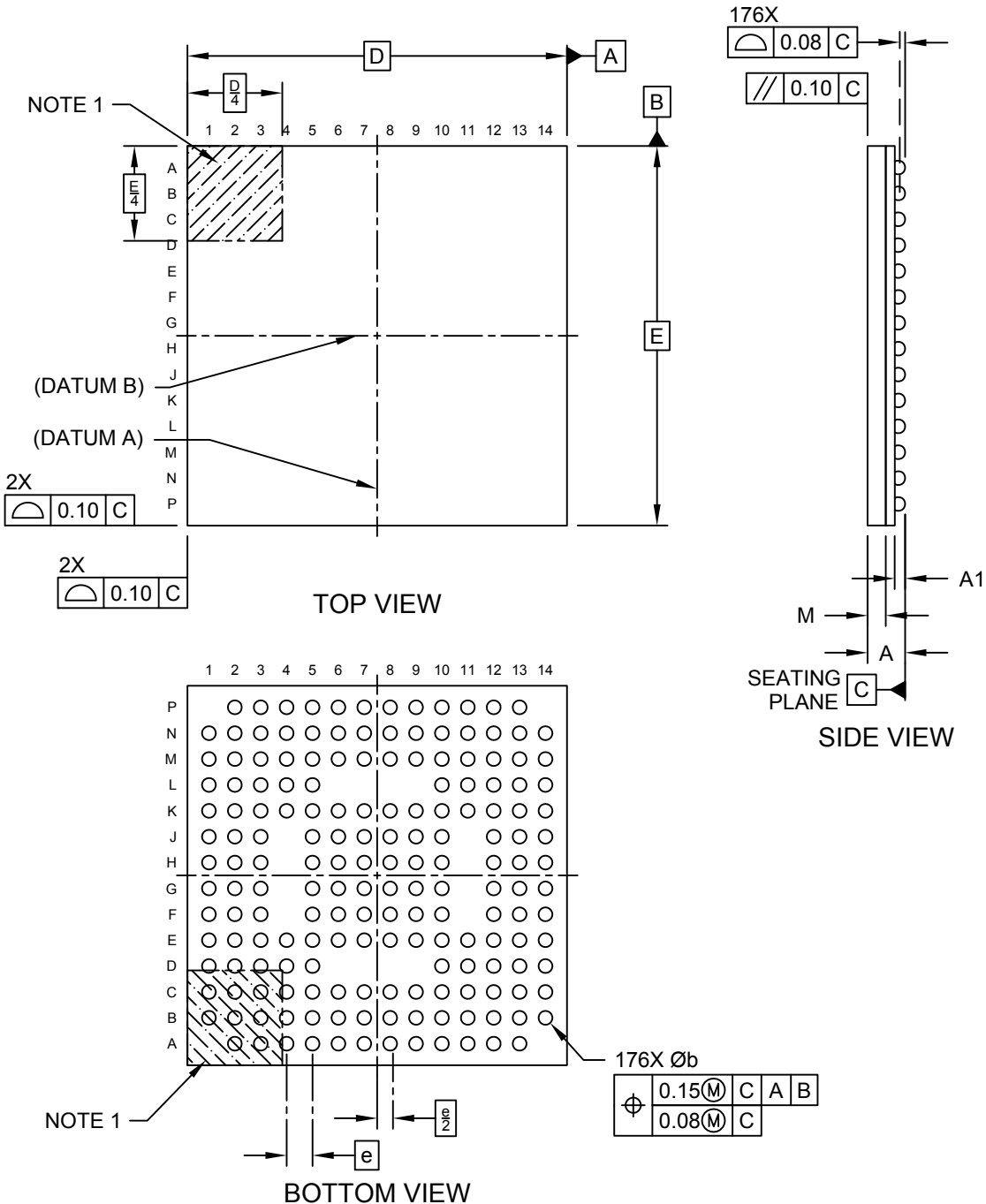


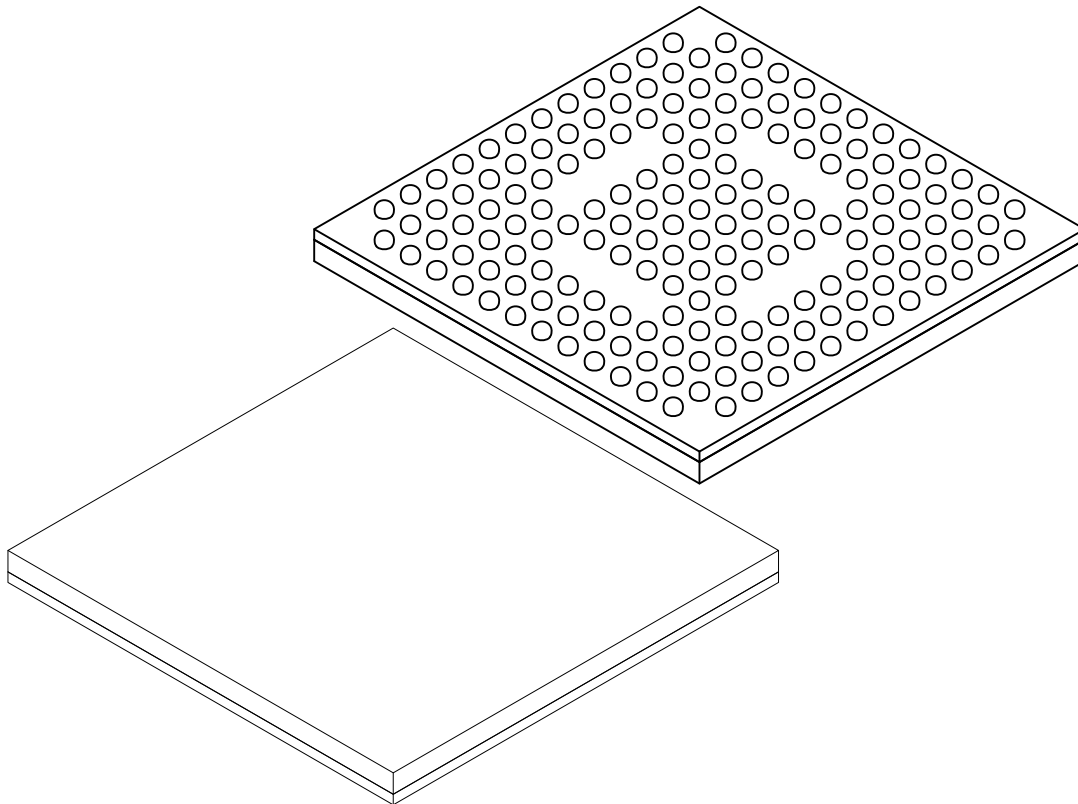
176-Ball Thin Fine Pitch Ball Grid Array (4LX) - 11x11 mm Body [TFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



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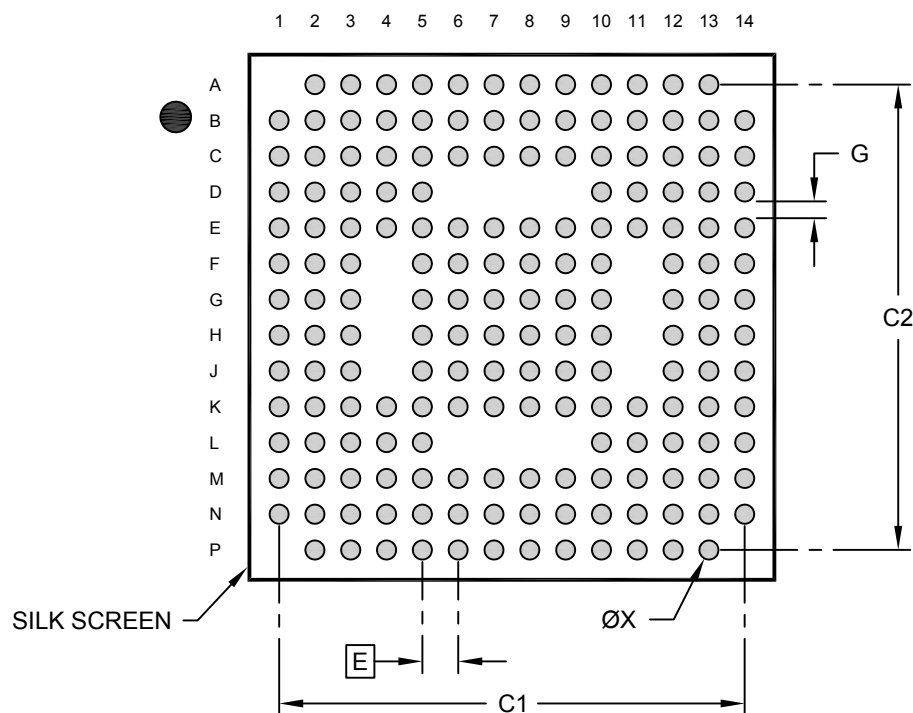
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	176		
Pitch	e	0.75 BSC		
Overall Height	A	-	-	1.19
Standoff	A1	0.21	0.30	-
Overall Length	D	11.00 BSC		
Overall Width	E	11.00 BSC		
Ball Diameter	b	0.35	0.40	0.45
Mold Cap Height	M	0.48	0.53	0.58

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

176-Ball Thin Fine Pitch Ball Grid Array (4LX) - 11x11 mm Body [TFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.75 BSC		
Contact Pad Spacing	C1		9.75	
Contact Pad Spacing	C2		9.75	
Contact Pad Diameter (X176)	X			0.40
Contact Pad to Contact Pad	G	0.35		

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-2481 Rev A